PRACTICAL-5

Aim:

To develop a rule-based expert system that identifies possible medical conditions based on user-provided symptoms and provides basic health advice using predefined rules.

Theory:

A **Rule-Based Expert System** is an artificial intelligence system that uses predefined rules and facts to make decisions or provide recommendations.
It consists of:

- **Knowledge Base:** Stores predefined rules about symptoms and possible conditions.
- **Inference Engine:** Matches user symptoms with the rules to identify the best match.
- **User Interface:** Accepts symptoms from the user and displays the result.

In this program, the system compares the user's symptoms with predefined rules for common cold, flu, allergy, migraine, and food poisoning. If at least two symptoms match a condition, it displays the possible condition and related advice.

Program/Code:

```
# Practical 5
# Rule-Based Expert System
# Problem: Simple Medical Diagnosis System

# Function to diagnose based on symptoms
def diagnose(symptoms):

    rules = [
        {
            "name": "Common Cold",
            "symptoms": {"fever", "cough", "runny nose", "sore throat"},
            "advice": "Take adequate rest, drink plenty of fluids, and monitor your symptoms."
        },
        {
            "name": "Flu",
            "symptoms": {"fever", "cough", "body pain", "headache", "fatigue"},
            "advice": "Take rest, stay hydrated, and consider consulting a doctor if symptoms are
severe."
        },
        {
            "name": "Allergy",
            "symptoms": {"sneezing", "runny nose", "itchy eyes"},
            "advice": "Avoid possible allergens and consult a doctor if symptoms persist."
        },
        {
            "name": "Migraine",
```

```
            "symptoms": {"headache", "nausea", "sensitivity to light"},
            "advice": "Rest in a quiet, dark room and consult a doctor if headaches are frequent or
severe."
        },
        {
            "name": "Food Poisoning",
            "symptoms": {"stomach pain", "vomiting", "diarrhea"},
            "advice": "Drink plenty of fluids and seek medical attention if symptoms are severe or
persistent."
        }
    ]

    best_match = None
    best_score = 0

    # Apply rules
    for rule in rules:
        matched_symptoms = symptoms.intersection(rule["symptoms"])
        score = len(matched_symptoms)

        if score > best_score:
            best_score = score
            best_match = rule

    # Display result
    if best_match is not None and best_score >= 2:
        print("\n====================================")
        print("         DIAGNOSIS RESULT")
        print("====================================")

        print("Possible Condition :", best_match["name"])
        print("Matching Symptoms  :", best_score)

        print("\nAdvice:")
        print(best_match["advice"])

        print("\nNote: This is a rule-based demonstration")
        print("and not a substitute for professional medical advice.")

    else:
        print("\nNo matching condition found.")
        print("Please consult a healthcare professional.")


# Main program
def main():
```

```
print("=====================================")
print("     RULE-BASED EXPERT SYSTEM")
print("       SIMPLE DIAGNOSIS SYSTEM")
print("=====================================")

print("\nAvailable symptoms:")

available_symptoms = [
    "fever",
    "cough",
    "runny nose",
    "sore throat",
    "body pain",
    "headache",
    "fatigue",
    "sneezing",
    "itchy eyes",
    "nausea",
    "sensitivity to light",
    "stomach pain",
    "vomiting",
    "diarrhea"
]

for i, symptom in enumerate(available_symptoms, 1):
    print(f"{i}. {symptom}")

print("\nEnter your symptoms one by one.")
print("Type 'done' when finished.\n")

symptoms = set()

while True:
    user_input = input("Enter symptom: ").lower().strip()

    if user_input == "done":
        break

    if user_input in available_symptoms:
        symptoms.add(user_input)
    else:
        print("Symptom not found. Please choose from the list.")

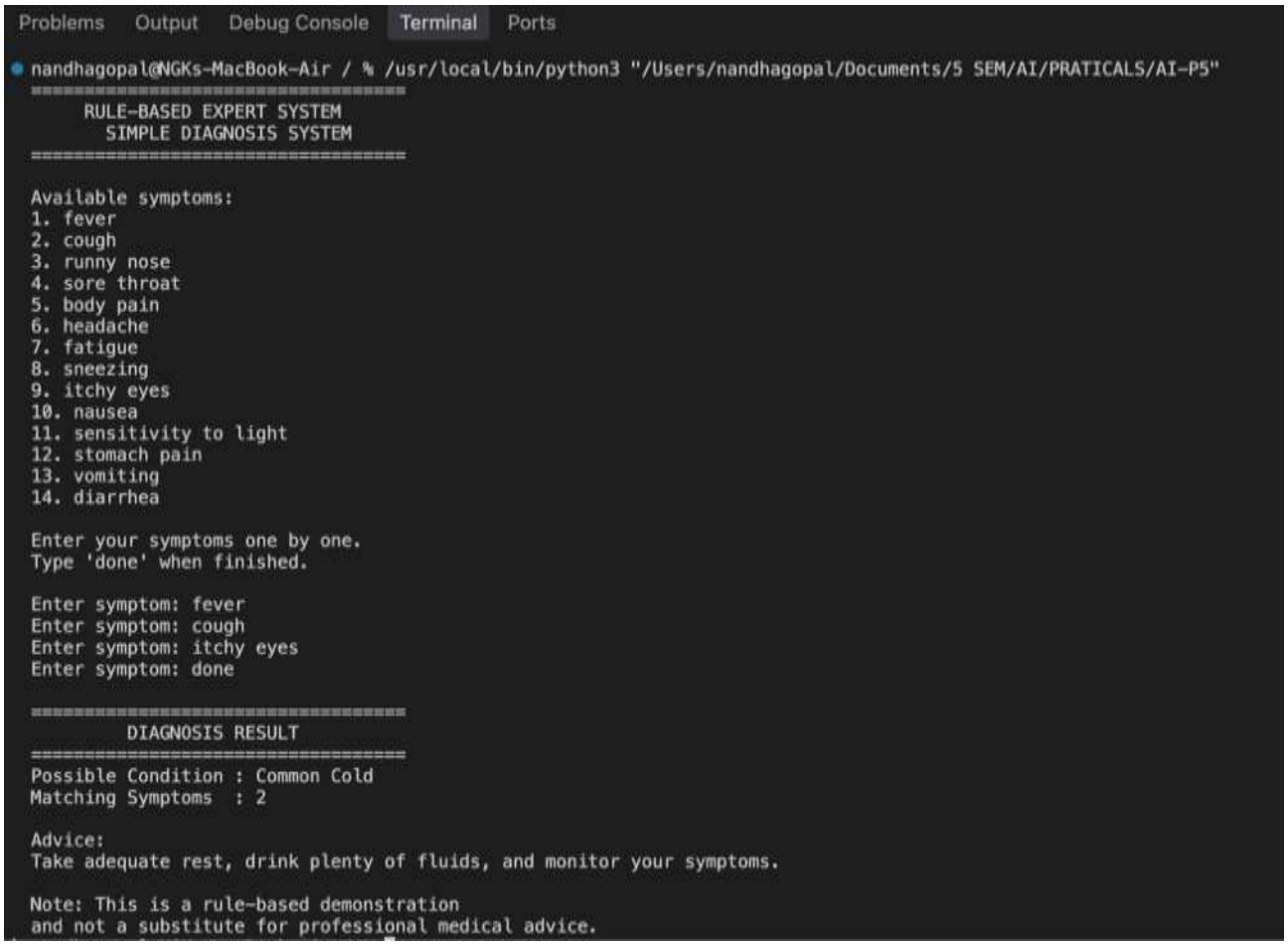
if len(symptoms) == 0:
    print("\nNo symptoms were entered.")
```

```
    return

# Run inference engine
diagnose(symptoms)


# Start the expert system
if __name__ == "__main__":
    main()
```

Output:



Conclusion:

Thus, a rule-based expert system was developed successfully to identify possible medical conditions based on user symptoms. The system uses predefined rules and an inference engine to match symptoms and provide basic advice. It demonstrates how artificial intelligence can be used for simple decision-making and advisory applications.